Name _____ Period _____ Date _____

3-7 Lesson Quiz: Transformations of Polynomial Functions

short form

2. Identify the parent function of the function $f(x) = -x^4 - 1$, and list the transformations from the parent function to create $f(x)$.

Ⓐ x^4; translate down 1 unit, reflect over the x-axis

Ⓑ x^4; reflect over the x-axis, translate down 1 unit

Ⓒ x^3; reflect over the x-axis, translate down 1 unit

Ⓓ x^3; compress the graph by a factor of x, translate down 1 unit

Explain your choice: why is the order of the two transformations in your answer the right order?

3. How does the graph of $g(x) = 3x^3 + 6$ differ from the graph of its parent function $f(x) = x^3$? Select **all** the transformations.

☐ **A.** The leading coefficient, 3, translates the graph up 3 units.

☐ **B.** The leading coefficient, 6, compresses the graph vertically.

☐ **C.** The leading coefficient, 3, stretches the graph vertically.

☐ **D.** Adding 6 translates the graph up 6 units.

☐ **E.** Adding 6 translates the graph right 6 units.

4. Determine the equation of the graph as it relates to either its parent cubic function or quartic function.

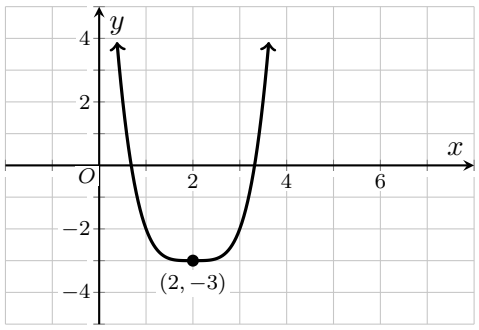
Ⓐ $f(x) = (x-2)^3 - 3$

Ⓑ $f(x) = (x-3)^4 - 2$

Ⓒ $f(x) = (x+3)^4 + 2$

Ⓓ $f(x) = (x-2)^4 - 3$

Explain your choice: how does the graph show whether the parent function is cubic or quartic?



Scoring: Question 2 and Question 3 are 10 points each. Question 4 is a bonus of up to 5 points. Total out of 20. (Numbers match the Savvas Lesson Quiz; Questions 1 and 5 are not on this form.)

3-7 Lesson Quiz, short form: answer key

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2. **B.** $-x^4 - 1$ is x^4 multiplied by -1 (reflection over the x-axis) and then 1 subtracted (translate down 1). Translating first and then reflecting gives $-(x^4 - 1) = -x^4 + 1$, which is why A is wrong; C and D have the wrong parent (the degree is 4). Savvas item 2, DOK 2, Realize skill F63. Taught by Example 3 (stretch before the vertical translation) and rehearsed by the packet Practice (TE Practice 18, $-x^4 - 8$, and Practice 8). **10 points:** 7 for B; 3 for the explanation (the -1 multiplies only x^4, so the reflection comes first; translating first would give $-(x^4 - 1) = -x^4 + 1$).

3. **C and D.** The leading coefficient 3 multiplies every y-value (vertical stretch); adding 6 moves every point up 6. A and B misread the coefficient as a translation or a compression; E confuses a vertical with a horizontal translation. Savvas item 3, DOK 2, Realize skill F63. Taught by Example 3A and Try It 3 ($2x^3 - 5$). **10 points:** 5 each for C and D, minus 2 for each wrong box checked (not below 0).

4. **D,** $f(x) = (x-2)^4 - 3$**.** Both ends point up and the graph turns once at its minimum, so it is the quartic $y = x^4$ with no reflection; the turning point $(0, 0)$ moved to $(2, -3)$: right 2, down 3. A is a cubic; B and C swap or misplace the translations. Savvas item 4, DOK 2, Realize skill F63. Taught by Example 4 (cut); answered by reading Example 3 backwards ($h = 2$, $k = -3$). **Bonus 5:** 3 for D; 2 for the explanation (both ends point up, as for $y = x^4$, so the parent is quartic).

Items left off this form (available for a retake or Form B): Savvas item 1, classify a graphed cubic as even, odd or neither (answer: odd; the graph is symmetric about the origin, crossing near $x = -1.7$, 0, 1.7; DOK 1, F63; Example 1, cut); Savvas item 5, cube volume in cubic centimeters $V(x) = x^3$ rewritten in cubic millimeters, $V(x) =$ _____ x^3 (answer: 1000, $V(x) = 1000x^3$; DOK 2, F63; Example 5, cut).

Graphs: the Question 4 graph is redrawn from the TE description (upward quartic, labeled minimum $(2, -3)$, window x from -2 to 8, y from -5 to 5) and the TE answer $f(x) = (x-2)^4 - 3$. The labeled point is part of the question, not the answer.

IXL replacement: SmartScore 70 or above on KT8 and RSN replaces this quiz with 20/20.